

Interaction of endothelial cell-selective adhesion molecule and MAGI-1 promotes mature cell–cell adhesion via activation of RhoA

Ritsuko Kimura¹, Tatsuro Ishida^{2*}, Masamitsu Kuriyama¹, Ken-ichi Hirata² and Yoshitake Hayashi¹

¹Division of Molecular Medicine & Medical Genetics, Department of Pathology, Kobe University Graduate School of Medicine, 7-5-1 Kusunoki-cho, Chuo-ku, Kobe 650-0017, Japan

²Division of Cardiovascular Medicine, Department of Internal Medicine, Kobe University Graduate School of Medicine, 7-5-1 Kusunoki-cho, Chuo-ku, Kobe 650-0017, Japan

Endothelial cell-selective adhesion molecule (ESAM) is a member of the immunoglobulin superfamily and mediates homophilic adhesion between endothelial cells. ESAM has been shown to bind to membrane-associated guanylate kinase (MAGUK) with inverted domain structure 1 (MAGI-1), but the interaction between these molecules remains unknown. We investigated the role of ESAM in the subcellular localization of MAGI-1 and cell adhesion by means of transfection experiments using Chinese hamster ovary (CHO) cells. Overexpression of ESAM recruited MAGI-1 to the cell–cell contact area. The intracellular domain of ESAM was necessary for the recruitment of MAGI-1 to the cell contact area, but did not participate in the initial cell–cell adhesion. Cell dissociation assays revealed that colocalization of ESAM and MAGI-1 promoted actin polymerization through the postsynaptic density 95/discs large/zonula occludens-1 (PDZ) domain and resulted in firm cell–cell adhesion, which was inhibited by an actin polymerization inhibitor. When the cells attach to each other, colocalization of ESAM and MAGI-1 can lead to the actin polymerization at intracellular contacts. Interaction of ESAM with MAGI-1 activated RhoA, and ESAM-mediated MAGI-1 recruitment to the cell membrane and mature cell adhesion were inhibited by a RhoA inhibitor. These findings suggest that ESAM may regulate MAGI-1 recruitment to the cell contacts, and subsequently promote actin polymerization and mature cell–cell adhesion through a RhoA-dependent mechanism.

Introduction

The vascular endothelium, located at the interface between tissues and blood, form the principal barrier by gating the traffic of molecules and cells across the vessel wall. The vascular endothelium plays crucial roles in the regulation of vascular permeability, vascular tone, leukocyte transmigration, thrombus formation, oxidative stress, and angiogenesis (Vestweber *et al.* 2009). Endothelial dysfunction leads to trigger the initiation and progression of a wide range of vascular diseases. The endothelial functions are known to largely depend on the

interaction between endothelial cells. Endothelial cells have tight and adherent junctions which are similar to that described for epithelial cells (Gumbiner 1996). The ability of the tight junction protein complex to form a seal is dependent on the interaction of the tight junction protein complex with structurally organized actin filaments in the cytoskeleton (Schneeberger & Lynch 2004). Well-known proteins constituting the tight junction complex include occludin, junctional adhesion molecule (JAM), claudin family members, and linker proteins such as ZO-1 that affiliate with the underlying actin cytoskeleton (Fanning *et al.* 1998).

We have recently expanded the repertoire of tight junction proteins by cloning of endothelial cell-selective adhesion molecule (ESAM), which belongs to

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*Correspondence: ishida@med.kobe-u.ac.jp

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the immunoglobulin superfamily (Hirata *et al.* 2001). ESAM is selectively expressed by vascular endothelial cells, platelets, and primitive hematopoietic progenitors (Hirata *et al.* 2001; Nasdala *et al.* 2002; Yokota *et al.* 2009). In endothelial cells, ESAM is localized at the tight junctions and mediates homophilic mechanical adhesion (Hirata *et al.* 2001; Nasdala *et al.* 2002). Previous cell culture- or animal studies have demonstrated that ESAM regulates tumor angiogenesis, basal- and vascular endothelial growth factor (VEGF)-induced vascular permeability (Ishida *et al.* 2003; Wegmann *et al.* 2006; Hara *et al.* 2009), as well as neutrophil extravasation (Wegmann *et al.* 2006; Khandoga *et al.* 2009). In addition, ESAM expression of tumor lymphatic endothelial cells was associated with lymph node metastasis of malignant tumors (Clasper *et al.* 2008). Recently, Rohatgi *et al.* (2009) reported that plasma levels of soluble ESAM are associated with subclinical atherosclerosis, major cardiovascular risk factors as well as with multiple inflammatory markers. These findings may recognize ESAM as a biomarker and modulator of angiogenesis, and atherosclerosis, or hematopoietic cell differentiation. However, these physiological actions of ESAM cannot be solely explained by its action on interendothelial interaction, and it is reasonable to raise a hypothesis that ESAM may interact with intracellular molecules and/or activate intracellular signaling cascade.

It has been demonstrated that the MAGUK family regulates tight junction assembly and cell adhesion by connecting transmembrane proteins with the cytoskeleton (Hirabayashi *et al.* 2003; Müller *et al.* 2003). MAGI-1 is a member of the MAGUK protein family, which is expressed at endothelial cell contacts and functions as a scaffold molecule (Wegmann *et al.* 2004). MAGI-1 is composed of an amino terminal guanylate kinase domain, six PDZ domains, and two WW domains flanked by the first and second PDZ domain (Dobrosotskaya *et al.* 1997). A previous study indicated that ESAM interacts with MAGI-1 through the PDZ domain (Wegmann *et al.* 2004). MAGI-1 can interact with a variety of intracellular molecules including such as mNET1 (Rho family nucleotide exchange factor) (Dobrosotskaya 2001), phosphatase and tensin homologue deleted from chromosome 10 (PTEN) (Chastre *et al.* 2009), or β -catenin (Dobrosotskaya & James 2000). However, little is known about the physiological significance of the interaction of ESAM and MAGI-1. To obtain better understanding of the function of ESAM, in this study, we aimed to characterize the action of ESAM on subcellular localization of MAGI-1 in cell-cell adhesion.

Results

ESAM recruits MAGI-1 to intercellular contacts through the PDZ-binding domain

To elucidate the tight junction-targeting mechanism of ESAM, we constructed three kinds of deletion mutants of human ESAM (Fig. 1A). The ESAMwt encodes full-length wild-type ESAM, whereas the ESAM Δ IC is a truncated ESAM which lacks its cytosolic domain, and the ESAM Δ C4 lacks its carboxy terminus PDZ domain. We have generated CHO cells constitutively expressing these ESAM mutants and analyzed the localization by immunofluorescence. Mock (pEGFP-C1 vector)-transfected CHO cells (control CHO), which stably express green fluorescent protein (GFP) in the cytosol, were used as a control. As shown in Fig. 1B, wild-type ESAM (CHO-ESAMwt) was localized at the cell-cell contact area, while GFP was distributed in the cytosol. Similarly, the truncated ESAM mutant ESAM Δ IC (CHO-ESAM Δ IC) was localized at cell-cell contacts, which indicates that the intracellular domain does not contribute to the targeting of ESAM to cell membrane.

Next, we investigated the role of ESAM in the subcellular localization of MAGI-1. We constructed double transfectants of MAGI-1 and full-length ESAM (CHO-MAGI-1/ESAMwt), MAGI-1, and ESAM Δ C4 in which PDZ-binding domain was deleted (CHO-MAGI-1/ESAM Δ C4), and MAGI-1 only (CHO-MAGI-1). The MAGI-1 expression was diffusely distributed in the cytosol in CHO-MAGI-1 (Fig. 1C). When MAGI-1 and ESAMwt were coexpressed (CHO-MAGI-1/ESAMwt), MAGI-1 expression turned concentrated at the cell contact and colocalized with ESAM. In contrast, when MAGI-1 and ESAM Δ C4 were cotransfected (CHO-MAGI-1/ESAM Δ C4), MAGI-1 expression remained in the cytosol. These results indicate that ESAM may recruit MAGI-1 to cell contact sites by interaction with the PDZ-binding domain. However, there was a weak expression of MAGI-1 at cell contact sites in CHO-MAGI-1/ESAM Δ C4, implicating the existence of regulatory domain(s) other than the PDZ-binding domain in ESAM that can recruit MAGI-1 to the cell contact area.

Extracellular domain of ESAM mediates initial cell-cell adhesion

ESAM has a potential to mediate cell aggregation through a homophilic molecular interaction (Hirata *et al.* 2001). To gain more insight into the supposed

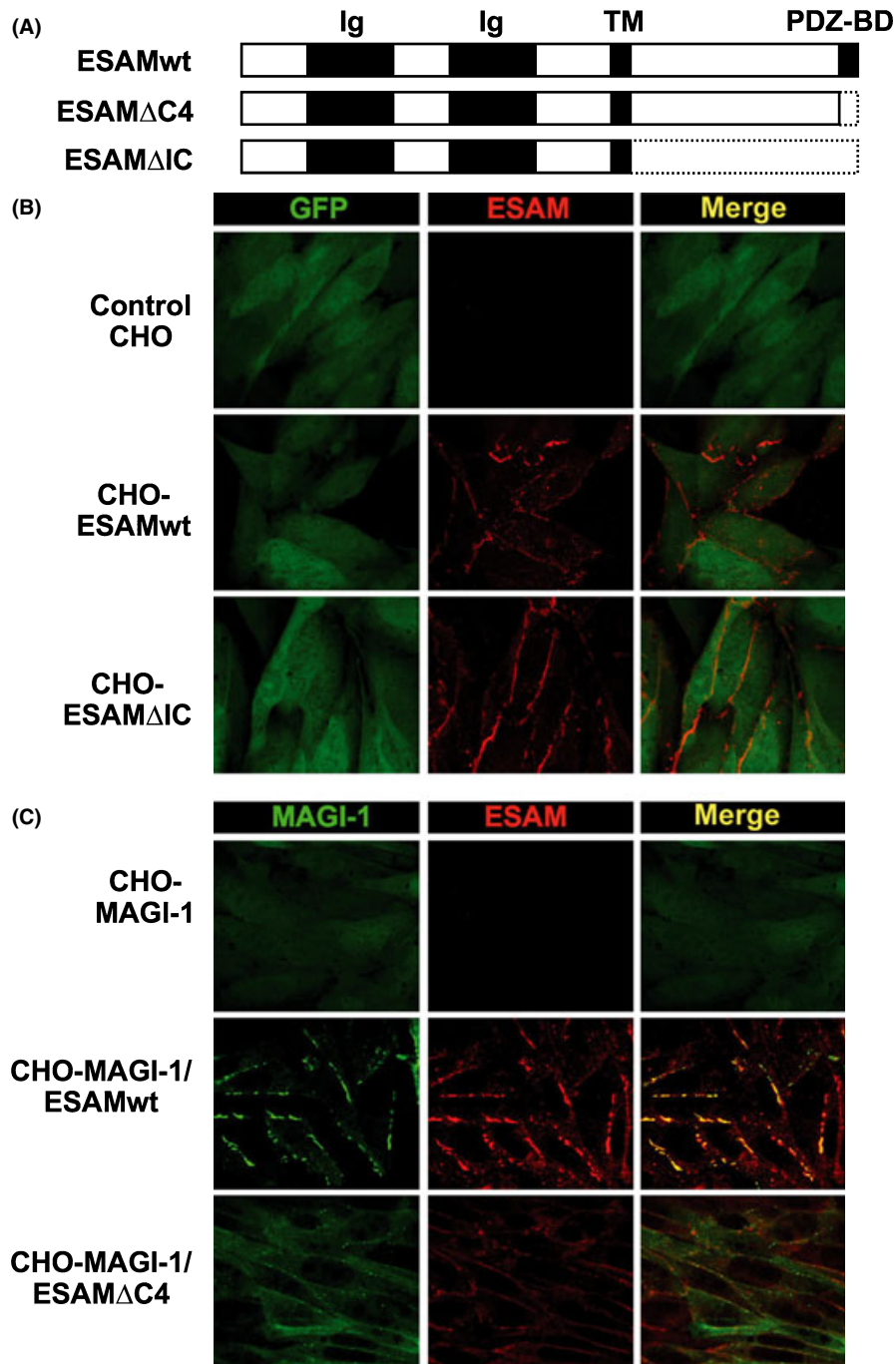


Figure 1 Intracellular domain of endothelial cell-selective adhesion molecule (ESAM) targets MAGI-1 to the cell membrane. (A) Deletion mutants of human ESAM. The ESAMwt encodes wild-type full-length ESAM, whereas the ESAM Δ IC encodes truncated ESAM which lacks its intracellular domain, and ESAM Δ C4 encodes truncated ESAM which lacks its PDZ-binding domain (PDZ-BD). CHO cells constitutively expressing these ESAM mutants and MAGI-1 were generated. Abbreviations: Ig, immunoglobulin domain; TM, transmembrane domain; PDZ-BD, PDZ-binding domain. (B) Full-length ESAM (CHO-ESAMwt) was localized at the cell–cell contact area, whereas green fluorescent protein was distributed in the cytosol. Truncated ESAM (CHO-ESAM Δ IC) was localized at cell–cell contacts. (C) Coexpression of ESAM and MAGI-1 (CHO-MAGI-1/ESAMwt) targeted these molecules to the cell membrane. In contrast, when the PDZ-binding domain of ESAM was deleted (CHO-MAGI-1/ESAM Δ C4), MAGI-1 expression was abundantly detected in the cytosol.

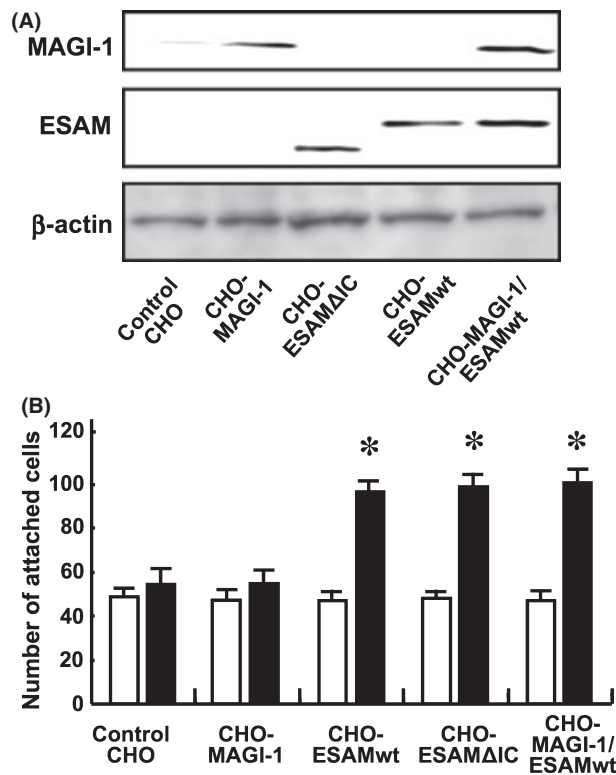


Figure 2 Extracellular domain of endothelial cell-selective adhesion molecule (ESAM) principally mediates initial cell adhesion. (A) Representative immunoblotting for the ESAM- or MAGI-1 expressions in stable CHO transfectants examined. (B) *In vitro* cell adhesion assay was performed for the assessment of initial cell-cell contacts using the stable transfectants and control Fc protein (open bars) and ESAM-Fc protein (solid bars). There were no differences in cell adhesion to control Fc protein among the groups. Whereas MAGI-1 expression did not affect the initial cell adhesion, expression of either ESAMwt (CHO-ESAMwt) or ESAMΔIC (CHO-ESAMΔIC) resulted in a comparable increase in cell adhesion compared to control CHO cells. Expression of MAGI-1 and ESAMwt (CHO-MAGI-1/ESAMwt) did not further augment the cell adhesion. * $P < 0.05$ versus control ($n = 3$).

function of ESAM, we evaluated the role of ESAM in initial cell contacts by means of cell adhesion assay. The stably transfected cell lines which have comparable expressions of ESAM and MAGI-1 were chosen (Fig. 2A) and plated onto culture plates coated either with ESAM-Fc- or with Fc protein (control). Control CHO or CHO-MAGI-1 did not show any adhesion activity to ESAM-Fc protein (Fig. 2B). In contrast, CHO-ESAMwt markedly adhered to the ESAM-Fc protein. Moreover, CHO-ESAMΔIC and CHO-MAGI-1/ESAMwt showed comparable adhesion activity as CHO-ESAMwt.

MAGI-1 contributes to ESAM-mediated mature cell-cell adhesion

To investigate the role of ESAM in the mature cell adhesion, we performed cell dissociation assays using the cell lines expressing ESAM or MAGI-1 at similar levels (Fig. 3A). Pipetting of the aggregates of control CHO and CHO-MAGI-1 turned them completely into single cell suspensions, while several cell aggregates

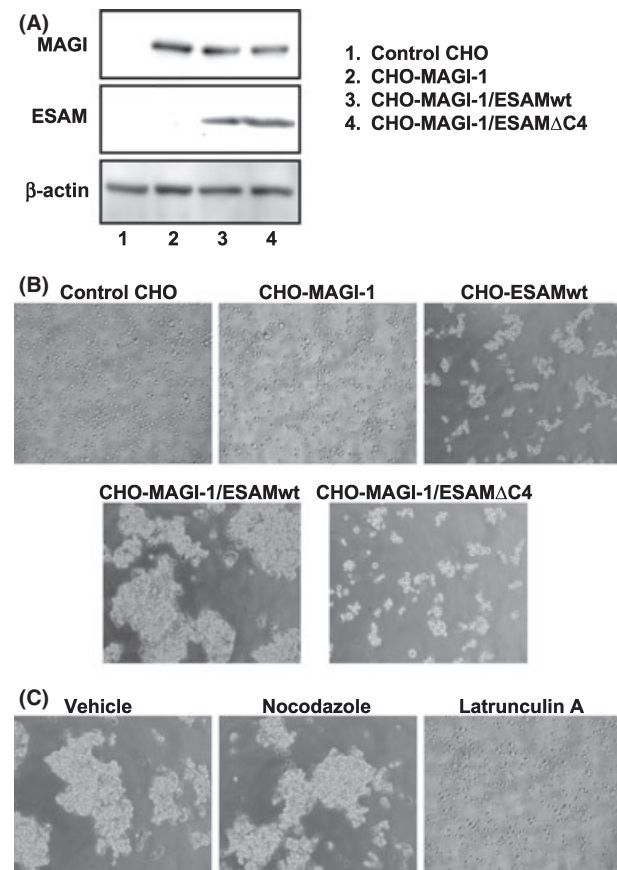


Figure 3 Interaction of endothelial cell-selective adhesion molecule (ESAM) and MAGI-1 promotes mature cell adhesion. (A) Representative immunoblotting for the ESAM- or MAGI-1 expressions in stable CHO transfectants examined. (B) Cell dissociation assays for the assessment of mature cell adhesion. Representative images from nine independent assays were shown. While control CHO and CHO-MAGI-1 became single cell suspension, CHO-ESAMwt exhibited cell aggregates after mechanical pipetting, and the cell aggregates were markedly observed in CHO-MAGI-1/ESAMwt cells. The deletion of ESAM intracellular domain (CHO-MAGI-1/ESAMΔC4) reduced the number of cell aggregates compared with CHO-MAGI-1/ESAMwt. (C) In CHO-MAGI-1/ESAMwt cells, the cell aggregation was abolished by the treatment of latrunculin A (an inhibitor of actin polymerization), but not by nocodazole (an inhibitor of microtubule polymerization).

remained in case with CHO-ESAMwt cells (Fig. 3B). In contrast, the same procedure hardly made CHO-MAGI-1/ESAMwt cells to single cell suspensions. The number of cell aggregates was less in CHO-MAGI-1/ESAM Δ C4 compared with CHO-MAGI-1/ESAMwt cells. Thus, MAGI-1 appeared to strengthen the cell–cell adhesion via PDZ-binding domain of ESAM, indicating that MAGI-1 promotes not initial but mature cell–cell adhesion mediated by ESAM.

Furthermore, we examined the effect of latrunculin A (an inhibitor of actin polymerization) or nocodazole (an inhibitor of microtubule polymerization) on the ESAM-mediated mature cell–cell adhesion. The cell dissociation assay revealed that latrunculin A treatment, which disrupts the actin cytoskeleton, made CHO-MAGI-1/ESAMwt cells dispersed (Fig. 3C), whereas nocodazole treatment had no effect on the cell dissociation. Thus, the ESAM/MAGI-1-mediated mature adhesion depends on the subsequent actin polymerization.

ESAM induces rearrangement of actin cytoskeleton via MAGI-1

Because tight junction is regulated by its affiliation with F-actin cytoskeleton (Miyoshi & Takai 2008), the distribution of actin filaments and MAGI-1 in confluent cells was detected by immunofluorescence. Control CHO cells showed a few actin filaments and stress fibers at the peripheral area of the cells (Fig. 4). In CHO-MAGI-1, the distribution of actin filaments was similar to that of control CHO cells. However, expression of ESAM (CHO-ESAMwt) resulted in a modest elongation of the cells as well as the aligned actin filaments in the cell contact area, compared to control CHO or CHO-MAGI-1. Moreover, CHO-MAGI-1/ESAMwt cells displayed colocalization of polymerized actin filaments and MAGI-1 accumulating along the cell margin. In CHO-MAGI-1/ESAM Δ C4, the signals for MAGI-1 as well as actin cytoskeleton along the cell margin were weaker than those in CHO-MAGI-1/ESAMwt. These results indicate that interaction of ESAM with MAGI-1 via the PDZ domain is likely to regulate the actin polymerization along the cell periphery, which presumably leads to the mature cell–cell adhesion.

Colocalization of MAGI-1 and ESAM at cell–cell contacts leads to actin polymerization

We compared serial changes in subcellular localization of ESAM, MAGI-1, and actin filaments, by means of

cell spreading assays using CHO-MAGI-1/ESAMwt. When the cell–cell contact was not yet formed within 3 h after incubation, MAGI-1 mainly displayed a cytoplasmic distribution (Fig. 5). However, at this time, ESAM and MAGI-1 began concentrating at the site of zipper-like early cell–cell contact. At 7 h, both MAGI-1 and ESAM were further abundantly concentrated at the extended cell contact area which was concomitant with the aligned actin filaments. At 12 h, both MAGI-1 and ESAM were predominantly localized at the mature cell contact area, where actin filaments were polymerized and existed along the cell–cell contact. These findings suggest that colocalization and interaction of ESAM and MAGI-1 at the cell–cell contact may lead to the actin polymerization.

RhoA is a major regulator for MAGI-1 localization and ESAM-mediated cell adhesion

The cytoskeleton structure is regulated by the Rho family small GTPases such as RhoA, Rac1, and CDC42 (Tapon & Hall 1997). It has been reported that knockdown of ESAM expression in endothelial cells resulted in an inhibition of Rho (Wegmann *et al.* 2006). On the other hand, MAGI-1 can bind to RapGEP (GDP/GTP exchange protein) (Mino *et al.* 2000) or mNET1 (Dobrosotskaya 2001). To determine whether small GTPases are implicated in ESAM-regulated actin rearrangement via MAGI-1, CHO-MAGI-1/ESAMwt cells were treated with a Rho inhibitor (Exo-C3) or a Rac inhibitor (NSC23766). Under the unstimulated condition, ESAM, MAGI-1, and actin filament were predominantly detected along the intercellular contact (Fig. 6A, top). Treatment of the cell with Exo-C3 markedly redistributed MAGI-1 expression to the cytosol with a decrease in ESAM expression on the membrane, which was accompanied by the reduced actin polymerization (Fig. 6A, bottom). In contrast, treatment of the cell with NSC23766 did not affect the accumulation of MAGI-1, while signals of ESAM and actin filaments were modestly decreased at the intercellular contacts. Furthermore, we compared the adhesion activity of these cells by the dissociation assay after treatment with Exo-C3 or NSC23766. Treatment of the cells with Exo-C3 resulted in more augmented cell dissociation than that with vehicle (Fig. 6B). In contrast, NSC23766 did not affect the cell dissociation when compared to the vehicle treatment. These findings indicate that RhoA is a major regulator of subcellular localization of MAGI-1 and the cell adhesion through actin polymerization.

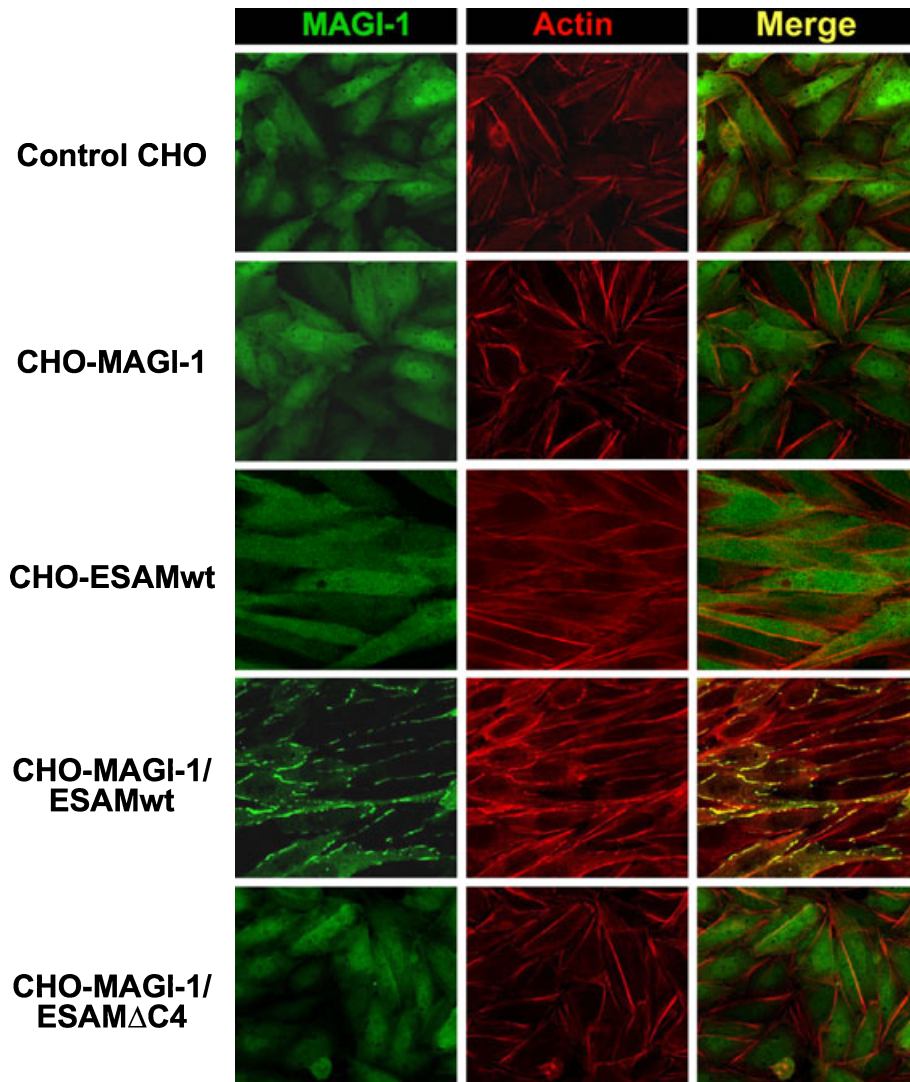


Figure 4 Colocalization of endothelial cell-selective adhesion molecule (ESAM) with MAGI-1 promotes actin polymerization. Immunohistochemistry for MAGI-1 and actin filaments. Control CHO or CHO-MAGI-1 cells showed a few actin filaments and stress fibers at the marginal area of the cell. In CHO-ESAMwt cells, the length of cell–cell contact area was extended, and actin filaments were aligned in the submembrane area. In CHO-MAGI-1/ESAMwt cells, MAGI-1 was accumulated along the cell margin, and strong signals of actin filaments were detected. CHO-MAGI-1/ESAM Δ C4 had weaker actin signals than that of CHO-MAGI-1/ESAMwt. Representative images from three independent assays are shown.

Because Exo-C3 promoted the redistribution of MAGI-1 and ESAM, we measured Rho activity in stable cells expressing ESAM or MAGI-1 by comparing the amount of GTP-bound form of RhoA. The Rho pull-down assay revealed that the level of activated RhoA was increased in CHO-MAGI-1 and CHO-ESAMwt cells compared to control CHO cells (Fig. 6C). The double transfectant CHO-MAGI-1/ESAMwt cells had a further increase in the activated RhoA level.

Discussion

Although the wide range of ESAM functions is considered to depend on its adhesion activity, the downstream of ESAM was not fully understood. This study has shown that extracellular domain of ESAM is required for the recruitment of ESAM to tight junctions and mediates the initial cell adhesion activity. In addition, ESAM contributes to the recruitment of MAGI-1 in intercellular contacts through the

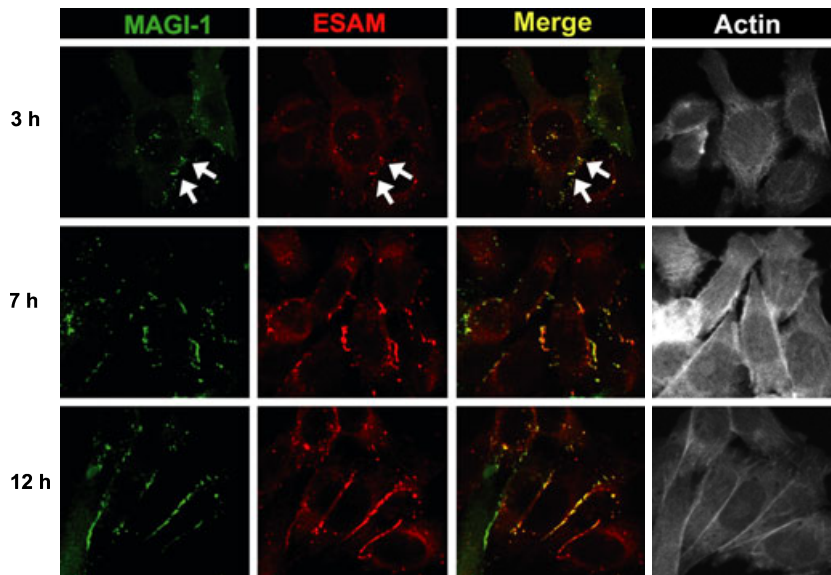


Figure 5 Time-dependent expression of MAGI-1 and actin polymerization. Time-dependent changes in subcellular localization of endothelial cell-selective adhesion molecule (ESAM), MAGI-1, and actin filaments were investigated by cell spreading assays using CHO-MAGI-1/ESAMwt. When the cell contact with the other cell at 3 h after incubation, ESAM and MAGI-1 began concentrating and colocalized at the site of zipper-like early cell–cell contact region (Arrows). At 7 h, both MAGI-1 and ESAM were abundantly concentrated at the extended cell contact area which was concomitant with the aligned actin filaments. At 12 h, both MAGI-1 and ESAM were predominantly localized at the mature cell contact area, where actin filaments were polymerized and existed along the cell–cell contact. Representative images from three independent assays are shown.

PDZ-binding domain. Given that an epithelial cell line lacking ZO-1 was shown to have delayed accumulation of claudin and occludin at the tight junction (Umeda *et al.* 2004), however, MAGI-1 may in turn contribute to the tight junction assembly and actin polymerization through interaction with ESAM. Moreover, we showed that, as the mature cell–cell adhesion was formed, ESAM and MAGI-1 were concentrated at the extended cell contact area where actin filaments were polymerized and existed along the cell–cell contact. It has been demonstrated that occludin and JAM colocalize with ZO-1 and E-cadherin in the spot-like adherens junctions prior to the formation of tight junction (Ando-Akatsuka *et al.* 1999; Ebnet *et al.* 2001). Therefore, it is considered that ESAM and MAGI-1 may be initially recruited into zipper-like early cell–cell adhesion and then concentrated at the extended cell–cell contact to promote mature cell adhesion by stabilizing actin polymerization. In this context, we have directly characterized the ESAM-mediated adhesion by combination of cell adhesion- and dissociation assays. The cell adhesion assay was used for the assessment of adhesive intensity of initial cell–cell contacts, whereas the cell dissociation assay was for assessment of adhesive intensity of mature cell–cell contacts (Carter *et al.*

1990). The ESAM-mediated cell–cell adhesion was strengthened by interaction with MAGI-1 through the PDZ-binding domain to achieve mature cell–cell adhesion, although MAGI-1 itself did not contribute to the ESAM-mediated initial cell–cell contact.

Previous studies have shown that knockdown of ESAM reduced the RhoA activation in cultured endothelial cells (Wegmann *et al.* 2006), whereas MAGI-1 binds to mNET1 (Dobrosotskaya 2001). In line with this, expression of ESAM and MAGI-1 resulted in an increase in RhoA activity. Moreover, in this study, inhibition of RhoA by Exo-C3 attenuated the accumulation of MAGI-1 and actin filament at cell–cell contacts, as well as adhesion activity mediated by ESAM/MAGI-1. The finding indicates that RhoA regulates maintenance of ESAM and MAGI-1 at tight junctions and in ESAM adhesion activity strengthened by MAGI-1.

The cytoskeletal reorganization of actin filaments is regulated by the Rho family GTPases (Tapon & Hall 1997; Chan *et al.* 2000). During formation of epithelial cell contacts, RhoA is activated and mediates expansion and completion of cell–cell adhesion (Yamada & Nelson 2007). In fact, activation of RhoA caused an accumulation of occludin and ZO-1 at cell junctions, whereas inactivation of RhoA suppresses

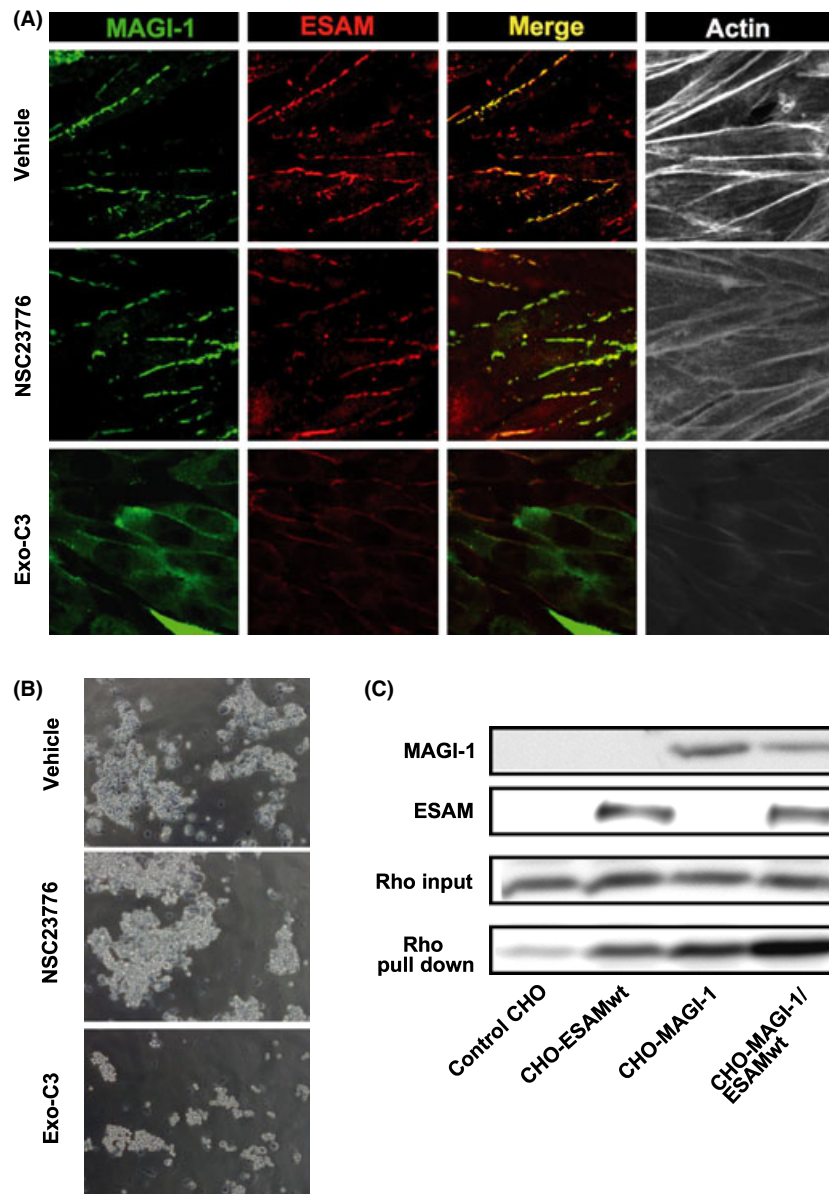


Figure 6 RhoA mainly regulates MAGI-1 localization and endothelial cell-selective adhesion molecule (ESAM)-mediated cell adhesion. (A) CHO-MAGI-1/ESAMwt cells were treated with a Rho inhibitor (Exo-C3, 1 $\mu\text{g}/\text{mL}$) or a Rac inhibitor (NSC23766, 100 μM) prior to immunohistochemical analysis. The NSC23766 treatment did not affect the subcellular localization of ESAM, MAGI-1, and actin filaments. The Exo-C3 treatment markedly redistributed MAGI-1 expression to the cytosol and reduced the actin polymerization. (B) Cell dissociation assay revealed that Exo-C3 resulted in more augmented cell dissociation than that with vehicle, while NSC23766 did not affect the cell dissociation. (C) The Rho pull-down assay revealed that the level of activated RhoA was increased by the overexpression of MAGI-1 (CHO-MAGI-1) and ESAM (CHO-ESAMwt) cells compared to control CHO cells. The double transfectant CHO-MAGI-1/ESAMwt cells exhibited a further increase in the activated RhoA level. Representative images from three independent assays are shown.

the expression of these proteins (Gopalakrishnan *et al.* 1998). RhoA activation promotes formation of stress fibers and focal adhesion complex, as well as phosphorylation of myosin light chain, which may inhibit

endothelial migration (Ren *et al.* 2000; Chang *et al.* 2009). In addition, sustained activation of Rho signaling increases the actomyosin assembly, cell adhesive interactions, and the expression of extracellular matrix

proteins and cytokines (Zhang *et al.* 2008). In contrast, there are some reports indicating that RhoA may promote endothelial migration. This study has demonstrated that interaction of ESAM and MAGI-1 activates RhoA, which resulted in actin polymerization and cell–cell adhesion. On the other hand, RhoA is known to control the activity of transcription factors such as NF- κ B, C/EBP, and serum response factor (Teusch *et al.* 2004). As far as we searched in the database, the ESAM promoter appears to contain NF- κ B/c-Rel-binding domains, suggesting that the ESAM function may be regulated by RhoA. Furthermore, it has been reported that increased RhoA activity and Rho kinase are associated with endothelial barrier functions (van Nieuw Amerongen *et al.* 2000; Wojciak-Stothard & Ridley 2003; Breslin *et al.* 2006; Sun *et al.* 2006; Kumar *et al.* 2009). Therefore, the physiological interaction between ESAM and RhoA activity should be further investigated using primary endothelial cells.

In summary, this study has characterized the role of ESAM in the subcellular localization of MAGI-1 and cell adhesion. The intracellular domain of ESAM contributes the recruitment of MAGI-1 to the cell–cell contact area through the PDZ domain-dependent mechanism. Colocalization of ESAM and MAGI-1 promoted actin polymerization and mature cell–cell adhesion, although ESAM-mediated initial cell–cell adhesion depends on the extracellular domain of ESAM. These actions were mediated by activation of RhoA. Thus, we have further expanded the mechanism of cell adhesion by clarifying the understanding of intracellular function ESAM.

Experimental procedures

Construction of wild-type and mutant ESAM

Full-length wild-type human ESAM (ESAMwt) were obtained by polymerase chain reactions using the forward primer 5'-ata-gaattcatgattccctcccggggc-3' and the reverse primer 5'-ttagaattctcataccagagagccagcttgac-3'. The carboxy-terminally truncated ESAM, in which four amino acids for the PDZ-binding domain is deleted (ESAM Δ C4), was obtained using the reverse primer 5'-ttagaattctcaagcttgactctggcagggc-3' and the above forward primer. The truncated ESAM which lacks its intracellular domain (ESAM Δ IC) was obtained using the reverse primer 5'-ttagaattctcacaaggaccagccagc-3' and the above forward primer. These cDNA fragments were cloned into the pCDNA3.1 vector (Invitrogen, Carlsbad, CA, USA). The pEGFP-C1-hMAGI1b and pEGFP-C1 were obtained from Dr N. Mochizuki (National Cardiovascular Center, Japan) (Sakurai *et al.* 2006).

Cell culture and transfection

CHO cells (American Type Culture Collection, Manassas, VA, USA) were transfected with pEGFP-C1 (mock) or pEGFP-C1-hMAGI-1b with a Lipofectamine and Lipofectamine Plus reagent (Invitrogen). Stable transfectants were selected with 800 μ g/mL G418 (Invitrogen). These transfectants were then transfected with pcDNA3-ESAMwt, pcDNA3-ESAM Δ C4, or pcDNA3-ESAM Δ IC, to generate double transfectants. Stable double transfectants were selected with 600 μ g/mL hygromycin B (Invitrogen). At least three clones in each stable transfectant were used for subsequent experiments. To specifically inhibit the activity of RhoA or Rac1, confluent CHO-MAGI-1b-ESAMwt cells were pretreated with 1 μ g/mL Clostridium botulinum C3 exoenzyme (Cytoskeleton, Denver, CO, USA), or 100 μ M NSC23766 (Rac1 inhibitor; Calbiochem, San Diego, CA, USA) for 16 h, followed by washing with phosphate-buffered saline (PBS). The cells were then subjected to the subsequent assays.

Immunofluorescence and immunoblotting

Cells were fixed with 4% formaldehyde in PBS and permeabilized with 0.1% Triton X-100 for 5 min. Cells were blocked with 3% bovine serum albumin (BSA) for 1 h and incubated with anti-ESAM antibody (R&D Systems, Minneapolis, MN, USA) diluted with immunoreaction enhancer solution (Can Get Signal immunostain; TOYOKO, Tokyo, Japan) for 1 h at room temperature. Immunopositive reaction was visualized with Alexa 555-labeled secondary antibodies and Alexa Fluor 635 phalloidin (Molecular Probes, Eugene, OR, USA). The images were obtained by the Zeiss LSM 510 confocal microscope (Carl-Zeiss Japan, Tokyo, Japan).

Immunoblotting was performed as described previously (Hirata *et al.* 2001). Anti-ESAM (R&D Systems), anti-MAGI-1b (a gift from Dr Mochizuki at the National Cardiovascular Center) (Sakurai *et al.* 2006), or anti-hemagglutinin (HA) antibodies (Roche Diagnostics K.K., Basel, Switzerland) were used as primary antibodies. Horseradish peroxidase-conjugated anti-goat IgG (Santa Cruz Biotechnology, Santa Cruz, CA, USA) and anti-rat IgG (GE Healthcare, Buckinghamshire, UK) were used as secondary antibodies.

Cell adhesion assay

Twenty-four-well plates were coated with 10- μ g recombinant ESAM-Fc chimeric protein or control Fc protein (R&D systems) in PBS for 3 h at 37 °C, followed by blocking with 3% BSA in PBS for 3 h at 37 °C. CHO cells constitutively expressing ESAMwt, ESAM Δ IC, or MAGI-1 were suspended in Hanks' Balanced Salt Solutions with 1 mM EGTA and plated (2.0×10^5) onto each ESAM-Fc- or Fc protein-coated well at 37 °C for 2 h. After washing with PBS three times to remove nonadherent cells, adherent cells were counted and revised by finding the remainder between ESAM-Fc- and Fc protein-coated wells. Data are indicated as the means $\pm$ SE of

three independent experiments on three clones of each kind of stable cells. The significance of the differences among the experimental groups was determined by a one-way analysis of variance followed by the Bonferroni test for multiple comparisons. The level of statistical significance was set at $P < 0.05$.

Cell dissociation assay

Cell dissociation assay was performed as described previously (Hirabayashi *et al.* 2003). Briefly, 10^6 cells were grown in 60-mm dishes to confluency, and the medium was removed. When indicated, cells were treated with dimethyl sulfoxide, 1 $\mu\text{g}/\text{mL}$ of nocodazole, or 5 μM latrunculin A in fresh medium for 60 min. After being washed with PBS, cells were scraped off by rotating the plate one round under a 3-cm-wide silicone blade cell scraper (Sumitomo Bakelite, Tokyo, Japan). Cells were subsequently dissociated in 500 μL of PBS by pipetting with disposable 1-mL plastic tips. At least three independent experiments for each stable transfectant were performed for evaluation of cell dissociation.

Cell spreading assay

Twenty-four-well plates were coated with 10 $\mu\text{g}/\text{mL}$ fibronectin in PBS for 10 h at 4 °C followed by blocking with 1% BSA in PBS for 1 h at 37 °C. CHO-MAGI-1/ESAMwt cells (1.0×10^4) were plated onto the plate on ice for 5 min, and then incubated at 37 °C for the indicated time point. After washing with PBS, adherent cells were subjected to immunocytochemistry for evaluation.

Pull-down assay for RhoA GTP

For RhoA pull-down assay, the plasmids containing HAx3-tagged human small GTPase RhoA wild type (pEF-BOS-RhoA-WT), constitutive active mutant of RhoA-G14V (pEF-BOS-RhoA-G14V), and the RhoGTPase-binding domain of human Dia (63–260 AA) (pGEX2T-Dia-GBD) were gifts from Dr T. Satoh at Division of Molecular Biology, Kobe University, Kobe, Japan (Mizushima & Nagata 1990; Ueda *et al.* 2008). Pull-down assays for the activated form of RhoA was carried out as described previously (Ueda *et al.* 2008). Briefly, CHO cells were transfected with pEF-BOS-HAx3-tagged RhoA-WT or RhoA-G14V and incubated for 12 h prior to the pull-down assay. Transfected cells were lysed in a lysis buffer [50-mM Tris-HCl (pH 7.4), 150-mM NaCl, 20-mM MgCl_2 , 0.5% (w/v) Nonidet P-40] and centrifuged at 15 000 g for 10 min at 4 °C. The cell lysates were incubated with glutathione-sepharose beads conjugated with GST-Dia-GBD (10 μg) for 30 min at 4 °C. Subsequently, glutathione-sepharose beads were washed three times with the lysis buffer. Precipitated RhoA was detected by immunoblotting using an anti-HA antibody. At least three independent experiments for each stable transfectant were performed for evaluation.

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